

# CSCI 1302-Spring 2022- Assignment #2

## Due 2/18/2022

This assignment needs to be submitted to me via D2L. To avoid a late penalty, it must be received by 11:59 pm on 2/18/2022. Technical problems with submitting the file or network problems will not be a valid excuse for a late submission. I would advise not waiting too late to submit your file. If you need to submit the assignment late, you can still do it through D2L.

### DO PAY ATTENTION TO ALL THE DETAILS LISTED BELOW

This programming assignment is a GUI program. It requires JFrame and JPanel to be used to complete the assignment. Any other method including JavaFx will not be accepted and will earn zero points.

A couple important notes/requirements:

- Make sure you submit the .java files. Once you compile the files it will create some additional compile files like the .class file. You may include the additional files, but I only need the source .java file.
- Name your driver file **Csci1302\_hw2.java** This file will contain the JFrame code and a main method.
- Name your 2<sup>nd</sup> file **DrawPoly.java** This file will contain a JPanel class for drawing the image/listeners/event handler methods etc.
- Make sure you include the proper white space. Each programming statement should be on its own line and indenting should be used to show “ownership”.
- Make sure you follow the identifier naming requirements and inline documentation laid out in the Programming Requirements/Restrictions module in D2L.
- Include a decent amount of comments inside each file for calculations and other notable code. You should have at least 8 comments per file in addition to the “header” comment block at the top of the program and the comment blocks for the methods.
- You may work together on the assignment or ask for assistance. You are NOT allowed to copy someone else’s work or have someone do part or all of the assignment for you.

Program requirements:

- 1) Your project will incorporate a single JFrame and a single JPanel
- 2) Your project requires you to create an interactive game. The theme is completely up to you. I shouldn’t see a lot of the same themes. Be creative.
- 3) The game must start with a splash screen that provides directions for the game and a start button. The game will not start until the user clicks on the start button with the left mouse button. The start button can be a JButton or another clickable item on the screen or it can just be a spot on the screen that you capture if clicked on. (For instance, if they click within a rectangular area inside the JPanel). It must be a spot/button in the window, it cannot be the entire window.
- 4) There must be some avatar that is manipulated/moved by the user. For example, it could be a car, spaceship, “human” character, or something else.
  - This avatar must be made up of between 4 and 10 polygons(inclusive-meaning you are allowed to use as few as 4). I highly recommend you use the program I supplied you to “map” your polygon points in the arrays.

- The avatar must also include something else from the Graphics class like an oval or rectangle
  - The avatar should have some sort of animation quality to give the illusion of movement. Examples would be rotating car wheels, or a characters legs moving in a walking motion
  - Much of this grade will rely on the quality of the avatar and its interactive qualities.
  - The avatar can be moved with the mouse and/or keyboard keys.
- 5) There must be some interactive component on the screen other than the avatar controlled by the user. For example, the old game Pac-Man has walls that don't allow the avatar to move through (of course there are ghosts as well in that game). Another example would be a car game that might need to avoid things in the road to avoid damage. This does not have to be a randomly controlled component, but that is a common tool in games(consider the tank program). Games get pretty boring if the same thing shows up in the same spot every time.
  - 6) The game must have a terminating feature, for instance too much damage on a car or a car running out of fuel/battery. It also shouldn't be too hard for a 50 year old man to lose too easily.
  - 7) After the game concludes, there should be another splash screen with a final score for the game.
  - 8) The program should terminate only after the user closes the window.

\*You can expand on the requirements as long as it doesn't conflict with any of set requirements. Example, you must use JFrames, which means you cannot use JavaFX. On the other hand, you could add sound to your program as that doesn't invalidate any of the requirements.

#### Some things I don't want you doing with your code:

- a) Do not use any break or continue statements. This leads to sloppy programming techniques. The exception to this is that you are allowed to use break statements inside a switch.
- b) Do not copy code from another source. I am aware there are plenty of copyable games out there online. Do not copy code from any source other than the examples I gave you. **You are allowed to copy my code and modify it if it helps.**
- c) Do not give variables or objects poor names. For example, if I was creating a variable to hold the size of the user string, I might call it string\_count, message\_length or count. I would not call it x, s\_ct or m\_lng. This also applies to loop control variables. Do not use a loop variable called i or x or lp but something like counter.
- d) If you struggle with one part of the assignment. Move on to complete another part of it. A partially complete assignment will earn at least some points.

#### Things to consider:

- a) A program that compiles and runs without errors will generally score better.
- b) Much of this assignment grade will be based on the quality of the graphics and the interactive quality of the program. Simply meeting the requirements without a lot of effort may earn a D.
- c) Originality counts here too. I don't want to see an exact duplicate of my code with a different set of polygons
- d) If you do copy my code, at a bare minimum, you should make sure your comments are appropriately changed and that variable names reflect your subject material. For instance, a variable called car\_speed should not be used if you have a human avatar that is walking.
- e) Have fun with this assignment. It's my favorite assignment of the semester as it allows you to be a lot more creative with it.